

White Paper

The Evolving Role of Real-World Evidence in Oncology HTA Decision-Making: Opportunities and Challenges

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Introduction

The use of Real-World Evidence (RWE) as a component of Health Technology Assessment (HTA) submissions has seen a marked increase in the last 5 years, reflecting the growing recognition of RWE's potential to enhance the understanding of health interventions in real-world settings, thereby informing more comprehensive and nuanced decision-making.^{1,2}

While Randomised Controlled Trials (RCTs) remain the gold standard for evidence generation, they can be challenging to conduct in niche oncology populations or in rare tumours and hampered by long timelines required to generate mature overall survival data or variability in Standards of Care (SoC) across markets. These limitations underscore the growing importance of RWE in supporting regulatory and HTA decision-making in oncology.

This report builds on IQVIA's Institute for Data Science 2022 report on Impact of RWE on HTA decision-making.² It examines the practical application of RWE in oncology, a therapy area characterized by high levels of innovation on the one hand and frequent challenges around evidence uncertainty and data gaps on the other, where RWE can provide meaningful value. The report includes case studies illustrating the use of RWE as external control arm in HTAs and offers actionable insights for Health Technology Developers (HTDs) on enhancing the credibility and strategic impact of RWE in HTA processes.

Guidelines for the use of RWE in regulatory decisions

Although the potential of RWE has long been acknowledged, recent years have marked a transition toward its practical application, with increasing emphasis on more formalised guidance for Real-World Data (RWD) collection (**Table 1**) and use of RWE to inform regulatory decisions (**Table 2**). Such guidance has helped improve the clarity of regulatory requirements for RWE generation and can potentially facilitate the design of RWE studies for HTA purposes as well.

Another emerging trend is the use of RWE collected under a regulatory mandate for HTA purposes. For example, DARWIN EU (**Table 1**) is also accessible to HTA bodies to generate RWE to inform decisions on the use, pricing and reimbursement of healthcare technologies, and has so far been used for two HTA-related pilot projects in oncology.³



Table 1. Efforts to regulate RWD collection

• The European Health Data Space Regulation (EHDS)⁴, which entered into force in March 2025, aims to promote a single market for health data, encompassing Electronic Health Records (EHR), medicine use, medical devices, and artificial intelligence (AI) • In the UK (not covered in the EHDS), several regulations are in place to protect patient data, including NHS England’s measures and broader UK data protection legislation^{5,6}
• DARWIN EU⁷, established by the European Medicines Agency (EMA) and the European Medicines Regulatory Network in 2022, generates RWE to support EMA committees and national regulators in the European Union (EU) in making more data-driven decisions on medicines
• The Medicines and Healthcare products Regulatory Agency (MHRA) Data strategy 2024-2027⁸ aims to develop the necessary infrastructure, architecture, and expertise to deliver timely access to RWD, which can in turn result in actionable RWE. By enhancing the collection and utilization of RWD, the MHRA seeks to foster innovation and improve public health outcomes through more informed regulatory decisions
• Heads of Medicines Agencies (HMA)-EMA RWD Catalogues⁹ contain metadata collected from various RWD sources and studies, intended to improve transparency, interoperability, and the discoverability of adequate data sources for generating RWE, helping regulators, pharmaceutical companies and researchers to identify and use such data
• NordicRWE¹⁰: The Nordic countries offer a rich source of health data through national registries, cohort studies, and Electronic Medical Records (EMR) systems. NordicRWE, a dedicated real-world evidence company, harnesses this unique data infrastructure to provide valuable insights and analyses that support drug development, regulatory requirements, and post-launch surveillance

Table 2. RWE in regulatory decision-making

	Key Food and Drug Administration (FDA) guidance documents include the 2016 21st Century Cures Act, ¹¹ which established the legal basis for use of RWE in regulatory decision-making, the 2018 Framework for FDA’s RWE Program, ¹² which outlines the FDA’s approach to implementing the Cures Act mandate, and the 2023 Considerations for the use of RWD and RWE to support regulatory decision-making for drug and biological products, ¹³ which offers detailed guidance on how drug developers can use RWD and RWE in submissions ¹⁴
	EMA began formally recognizing the potential of RWE in its 2021 Vision for Use of RWE ¹⁵ , which emphasised data quality, interoperability and standardization. Its recent 2025 Reflection paper ¹⁶ offers critical guidance on the design, conduct and analysis of non-interventional studies using RWD to generate RWE for regulatory purposes
	MHRA outlined the guidance on the use of RWD in clinical studies to support regulatory decisions ¹⁷ (2021) and launched a Real-World Evidence Scientific Dialogue Programme ¹⁸ in March 2024 as a part of its Data Strategy, ¹⁹ which sets out the regulator’s vision for how data and digital technology will deliver improvements and drive innovation in healthcare

RWE in HTA decision-making

With the increasing utilization of RWE, HTA bodies are recognizing its importance in the evaluation of new technologies. RWE is increasingly accepted by HTA bodies to complement or supplement RCT data, especially where RCTs are unfeasible or limited. Several agencies have published guidelines for RWE use in HTA.

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The **National Institute for Health and Care Excellence (NICE) Real-World Evidence Framework (2022)**²⁰ emphasises three core principles: ensuring high-quality and relevant data, maintaining transparency and integrity throughout the study process, and using robust analytical methods to minimise bias and characterise uncertainty. The NICE framework is a 'living' document, which aims to adapt to advances in data, methods and tools through modular updates, with the most recent one from January 2025, which included a dedicated appendix focused on planning, conducting, and reporting of qualitative studies to support NICE decision making.

The **French National Authority for Health (HAS) Methodological Guide on Real-World Studies (2021)**²¹ provides practical guidance to improve quality, transparency, and reliability of RWE studies by offering benchmarks on study design, data sources, and analytical methods. The guide focuses on observational studies and pragmatic trials, particularly those required post-registration, and encourages alignment with international standards to enhance confidence in RWE findings.

The **2023 revised Transparency Committee doctrine**²² introduces the possibility of submitting

non-RCT data, including SATs with ECA or other forms of indirect comparison, as well as RWE, provided that it allows for comparison with available treatments, with the objective of facilitating reimbursement for products with immature evidence while maintaining an acceptable level of quality. The updated policy thus allows for indirect comparisons of good methodological quality, as well as data from control groups, provided they are explained and justified beforehand by the HTD.

The **Institute for Quality and Efficiency in Healthcare (IQWiG) General Methods 7.0 (2023)**²³ established a cautious foundation for the use of RWE in HTAs, but limited to specific contexts, such as when RCTs are unfeasible or unethical. Emphasis is placed on high data quality, transparency, and methodological rigour. IQWiG stresses the need to minimise bias and ensure comparability when using non-randomised data.

The **2025 IQWiG General Methods 8.0**²⁴ marked a shift toward greater formalization and clarity for the use of RWE in HTA. It highlighted a methodological emphasis on clarity, transparency and scientific rigour. IQWiG acknowledges the value of routine practice data collection, i.e. 'Anwendungsbegleitende Datenerhebungen (AbD)', which is the formal post-approval data collection scheme in Germany. However, such data must meet strict quality standards, including clear study design and methodology, bias control (e.g., confounding) and transparency in data source selection and analysis. Registries are highlighted as suitable sources for AbD studies if they provide high-quality, complete and relevant data. Outside of the General Methods, IQWiG has also continued to emphasise randomised registry studies as a viable solution.

The **IQWiG preliminary report outlines a scientific framework for the collection and analysis of routine practice data where there is limited clinical evidence available at market entry**²⁵ and emphasises the target trial emulation as the preferred design to minimise bias for studies using non-randomised evidence for causal inference purposes. It further introduces a systematic approach to confounder

identification, drawing on literature reviews, clinician input, and expert validation, as detailed in the **IQWiG working paper**.²⁶ The report also provides a quantitative framework for sample size estimation under uncertainty, offers recommendations for setting the correct observation start date to avoid immortal time bias, addresses treatment switching through both Intention-To-Treat (ITT) and pragmatic strategies, including bridging therapies, and provides recommendations for the interpretation of effect sizes that accounts for potential residual confounding (i.e. shifted null hypothesis testing). Finally, it stresses the importance of handling missing data through multiple imputation and supports the use of propensity score methods in small cohorts, provided confounders are prioritised to ensure model convergence.

2023 Guidance for Reporting Real-World Evidence,²⁷ published by Canada's Drug Agency (CDA-AMC, previously known as Canadian Agency for Drugs and Technologies in Health (CADTH)), Health Canada, and National Institute of Excellence in Health and Social Services (INESSS), lays the foundation for the use of RWE in regulatory and HTA decision-making in Canada, emphasizing fit-for-purpose use and methodological rigour. The guidance outlines a structured checklist for reporting RWE studies, which can be used as a tool to ensure submissions adhere to the guidelines outlined in the document. A separate **Response to Stakeholder Feedback report**²⁸ outlines the feedback received during the consultation process by theme and responses to feedback, including from regulatory and HTA perspectives.



Table 3. Guidance for use of RWE in HTA decision-making

	 NICE	 HAS	 IQWiG	 CDA-AMC
Guidance documents	NICE Real-World Evidence Framework ²⁰ June 2022	Real-world studies for the assessment of medicinal products and medical devices, Methodological guide ²¹ June 2021 Transparency Committee doctrine update ²² February 2023	The 2025 IQWiG General Methods 8.0 ²⁴ February 2025	Guidance for Reporting Real-World Evidence ²⁷ May 2023
Use of RWE	Accepted to complement RCTs, especially for long-term outcomes and rare diseases	Accepted to complement RCTs, especially for long-term outcomes and rare diseases. Indirect comparison data of good methodological quality allowed, as well as data from control groups, provided they are explained and justified beforehand by the HTD	Accepted under the term “versorgungsnaher Daten” (routine care data), particularly in the context of post-approval data collection (AbD), provided strict quality and methodological standards are met	Accepted to complement RCTs for regulatory and HTA decisions, especially where RCTs are infeasible or limited.
Requirements	Emphasis on data relevance, transparency, and robust analytical methods	Clear study design, reliable data sources, and alignment with international standards	Clear study design (e.g., observational, non-interventional), transparent data sources, robust bias control (especially confounding)	Clear study design, reporting and statistical methods; emphasis on transparency, reproducibility and fit-for-purpose use
Use cases	• <i>Characterising health conditions, interventions, care pathways, and patient outcomes, including natural history</i> • <i>Estimating economic burden and developing economic models</i> • <i>Measuring patient experience</i> • <i>Developing digital health technologies including prognostic models</i> • <i>Characterising health inequalities</i> • <i>Estimating test accuracy, e.g., biomarkers</i> • <i>Estimating device or procedure failure rates:</i> • <i>Measuring the impact of interventions, service delivery and decisions about care</i>	• <i>Describing the conditions of use, target and beneficiary population</i> • <i>Estimating utility scores</i> • <i>Measuring efficacy and risks related to the use of health products</i> • <i>Measuring the consumption of resources</i> • <i>Measuring the organizational impact</i>	• <i>Routine practice data collection (AbD) for benefit assessments</i> • <i>Supplementing RCTs where evidence gaps exist</i> • <i>Informing HTA decisions when RCTs are infeasible or unethical</i> • <i>Supporting long-term effectiveness and safety evaluations</i> • <i>Providing contextual data on treatment patterns and patient populations</i>	• <i>Rare diseases</i> • <i>Paediatric or elderly populations</i> • <i>Cases where clinical trials are unethical</i> • <i>Post-marketing evaluations</i> • <i>Long-term safety or effectiveness monitoring</i>

RWE and joint clinical assessment

Joint Clinical Assessment (JCA), which was recently introduced under the EU HTA Regulation,²⁹ sets out harmonised requirements for evaluating health technologies across EU Member States. The Methodological and Practical Guidance (MPG) subgroup has defined the evidence expectations for addressing the PICO (Population, Intervention, Comparator, Outcomes) in JCA dossiers.³⁰

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RWE can play a valuable role in hypothesis generation, assessing trial feasibility, informing Bayesian models, identifying patient baseline characteristics or prognostic and predictive factors, describing the use of a health technology in a real-world setting, assessing the effectiveness and/or safety of health technologies (e.g., for new indications of already-used technologies or for documenting long term follow-up), and supplementing or replacing control arms (e.g., in rare diseases).

However, the guidance also highlights the methodological challenges of using RWE, including confounding bias, measurement errors, and inconsistent follow-up due to its non-experimental origin. Anchored comparisons using Matching-

Adjusted Indirect Comparisons (MAICs) and Simulated Treatment Comparisons (STCs) are recommended, whereas unanchored approaches should generally be considered only with access to full Individual Patient Data (IPD) from all studies, due to the high risk of bias. The guidance emphasises that RWD is not a study design itself and that the internal validity of RWE depends heavily on the robustness of the study design and analytical methods used. Proper reporting on data sources, study design, and potential biases is essential to ensure the reliability of findings derived from RWD.

While it remains to be seen how RWE will be used and evaluated in the first JCA reports,³¹ the potentially high number of PICO³² could result in substantial evidence requirements, which cannot be fully addressed by clinical trial data. In such cases, the use of RWE is likely to become essential to support the assessment.

RWE and international collaborations

In parallel with the development of the JCA, several international HTA collaborations have emerged to explore methodological alignment and innovation. Notably, a multi-agency collaboration³³ was launched in 2022 involving Australia (Pharmaceutical Benefits Advisory Committee (PBAC)), Canada (CDA-AMC), the UK (NICE, Scottish Medicines Consortium (SMC), and Health Technology Wales (HTW)), with New Zealand (Pharmaceutical Management Agency (PHARMAC)) and Quebec (INESSS) joining in 2023. Additionally, the **Health Economics Methods Advisory (HEMA)**³⁴ group, established in 2025 by the Institute for Clinical and Economic Review (ICER), NICE, and CDA-AMC, seeks to critically and independently evaluate emerging methods and processes in HTA. While these collaborations aim to harmonise approaches and share best practices, they have not yet published formal guidance on the use of RWE.

The **Support Utilisation of Sustainable and Tailored INnovative methods for HTA (SUSTAIN)**³⁵ is an EU-wide initiative to build a supporting infrastructure to ensure ongoing implementation of the latest and fit-for-purpose HTA methodologies and tools in practice.

RWE4Decisions³⁶ is a multi-stakeholder learning network and initiative focused on improving the use of RWE to inform HTA and payer decisions across the lifecycle of highly innovative medicines. The initiative includes HTA authorities and payers, regulators, clinicians, academics, patient groups, pharmaceutical industry, data experts and disease registry holders.

The **GREG initiative (Global Real-world Evidence Collaboration)**³⁷ fosters international dialogue on best practices for RWE generation and application for regulatory and HTA decision-making.

The **GetReal Institute**³⁸ is a multi-stakeholder platform — of which IQVIA is a current member — that is dedicated to improving the quality, transparency, and acceptance of RWE in regulatory and HTA context.

Current challenges around the use of RWE in HTA

RWE methodologies have advanced significantly in recent years, with increasingly sophisticated data collection, analytics, and integration techniques. These innovations have enhanced the robustness and relevance of RWE, particularly in complex therapeutic areas such as oncology. However, the relevance of RWE in HTA decision-making has not progressed at the same pace. While the theoretical potential of RWE is widely acknowledged, its practical application in HTA remains inconsistent across countries.

Here we explore the challenges and opportunities of leveraging RWE in oncology HTA submissions across a range of RWE purposes, and with a focus on external control arms. We examine key barriers to adoption, such as methodological scepticism, concerns around data quality, and variability in acceptance across HTA bodies. We outline practical considerations for HTDs to enhance the credibility and impact of RWE in submissions with the aim of better harnessing RWE to support evidence-based decision-making, accurately capture the clinical benefit of novel technologies, and ultimately improve patient outcomes.

Application of RWE in Oncology

To evaluate the practical application of RWE in oncology, we conducted a targeted analysis of HTA submissions published between January 2021 and October 2024. Oncology was selected due to its high volume of HTA submissions, and the frequent presence of evidence gaps where RWE can add value (e.g., in rare indications or for long-term outcomes). Submissions were excluded if they lacked reference to RWE or did not contain statements derived from RWD. The dataset focused on HTA submissions in multiple myeloma and non-small cell lung cancer (NSCLC) (N=602). These two indications were selected due to the frequent use of RWE to address challenges in trial design, comparator selection, and evidence generation in complex or rare patient populations.

The analysis focused on assessing the contribution of RWE on HTA outcomes (**Figure 1**), examining whether and how RWE influenced final decisions across HTA bodies in Belgium, Denmark, Finland, France, Germany, the Netherlands, Norway, Spain, Sweden, and the United Kingdom. Only RWE derived from actual patient data in clinical settings was considered; extrapolated epidemiological estimates from clinical trials were excluded.

In addition to the above analysis, we reviewed specific case studies of RWE use in HTAs to extract actionable insights. These case studies highlight both the strengths and limitations of RWE in HTA submissions, offering practical guidance for enhancing the credibility and strategic value of RWE in future evaluations.

Figure 1. RWE contribution to HTA outcomes – category definitions

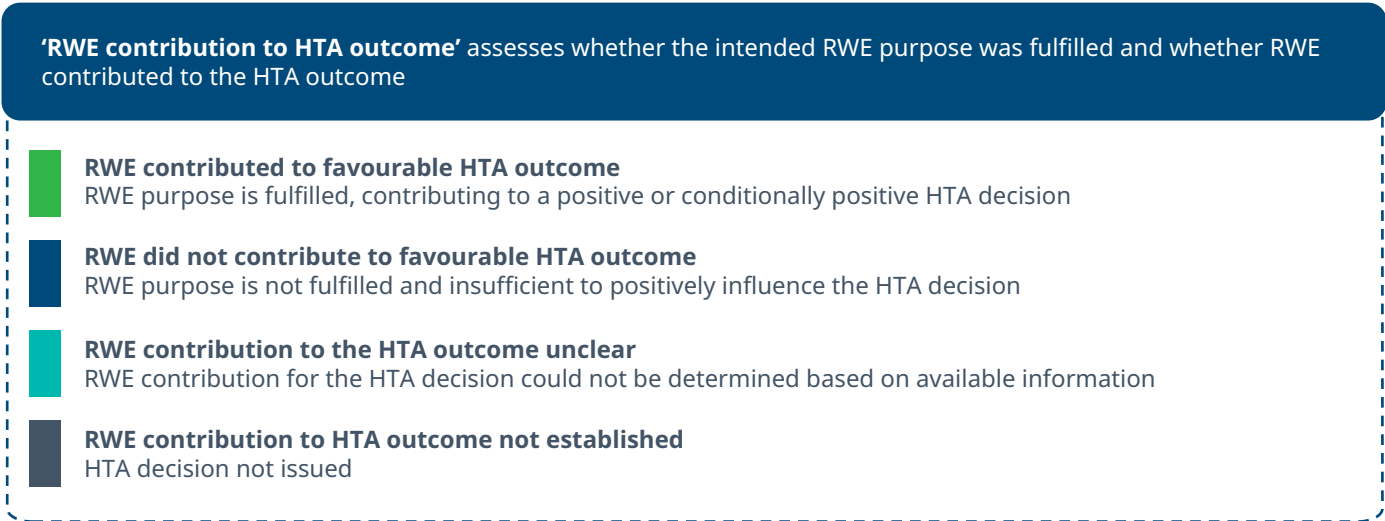
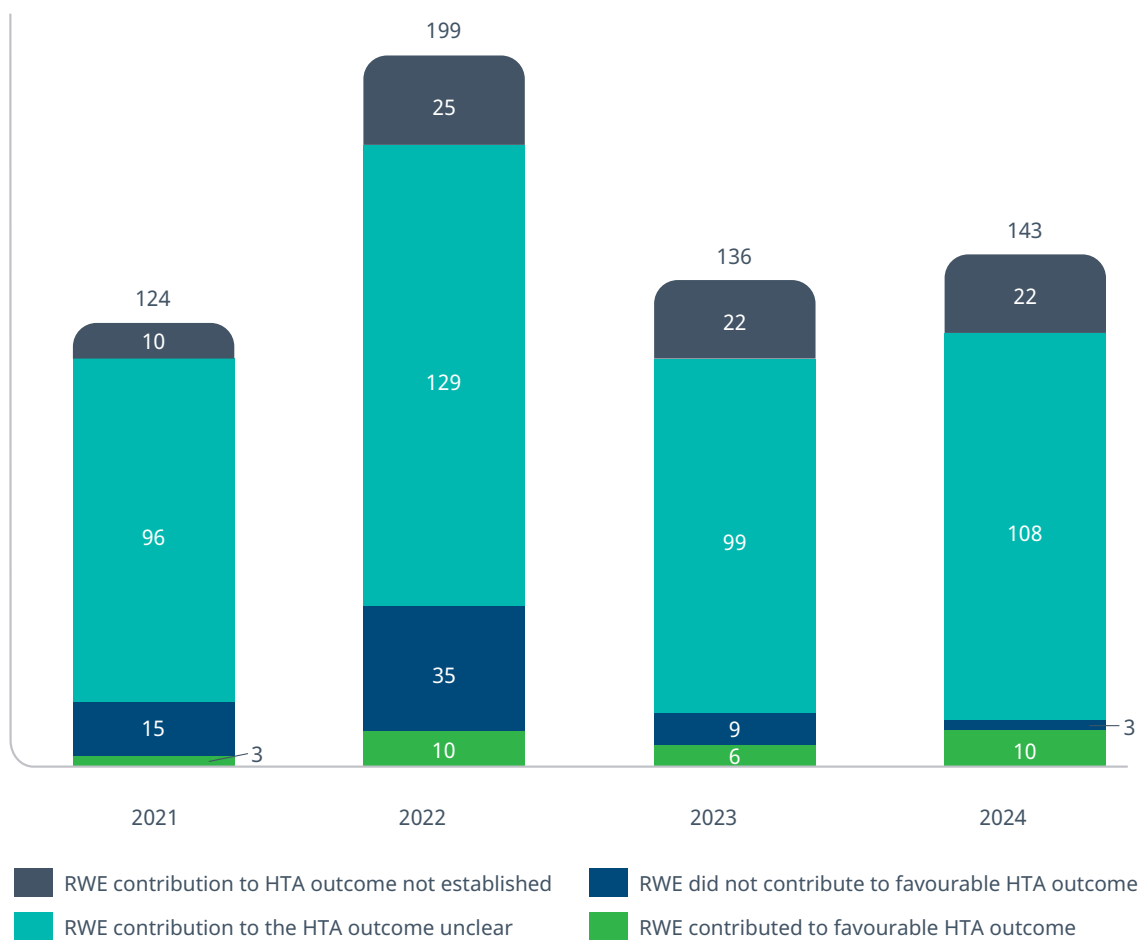


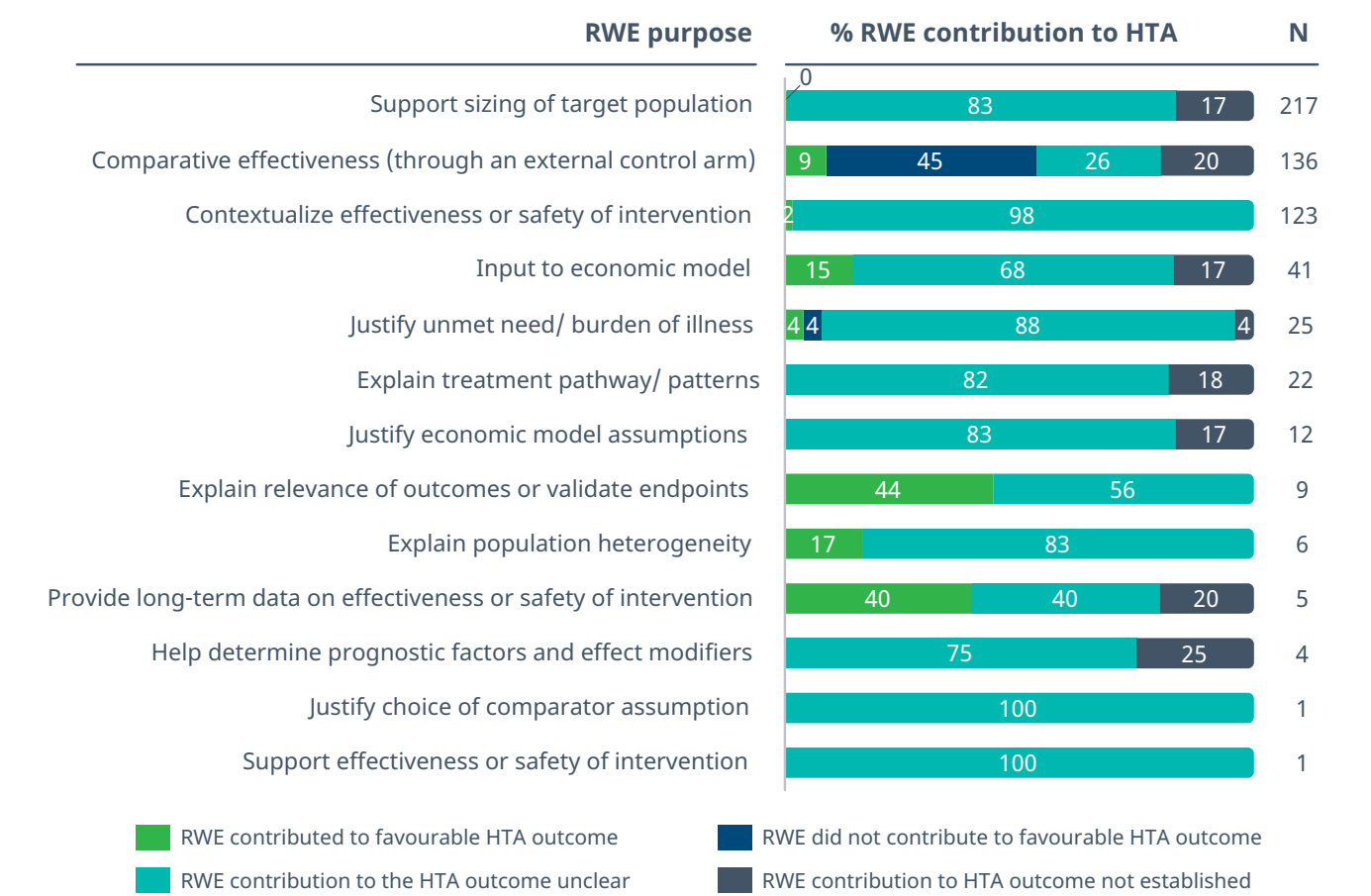
Figure 2. RWE contribution to HTA outcome (2021-2024, N=602 HTA submissions in multiple myeloma and NSCLC)



RWE contribution to HTA outcomes

While the number of HTAs with RWE has increased over the years,¹ RWE contribution to HTA outcomes has fluctuated over the years, with a substantial number of RWE sources (432 or 71.7%) associated with an unclear contribution to HTA outcomes (**Figure 2**).

Figure 3. RWE purpose and contribution to HTA outcomes (2021-2024)



Cases where RWE contributed to favourable HTA decision are seen for the use cases of explaining relevance of outcomes or validating endpoints, explaining population heterogeneity, providing long-term data on effectiveness or safety, and input to economic models (**Figure 3**).

A large proportion of assessments where use of RWE did not contribute to favourable HTA outcomes (45%) is related to the use case of providing comparative

effectiveness by means of an ECA. This reflects the heightened scrutiny HTA bodies apply when RWE is used in the context of comparative clinical effectiveness due to concerns around bias and confounding. Importantly, in many of these cases, HTA outcomes were likely to have been negative irrespective of the availability of RWE, often due to the absence of head-to-head clinical trial evidence, suggesting that the inclusion of RWE did not alter the overall decision trajectory.

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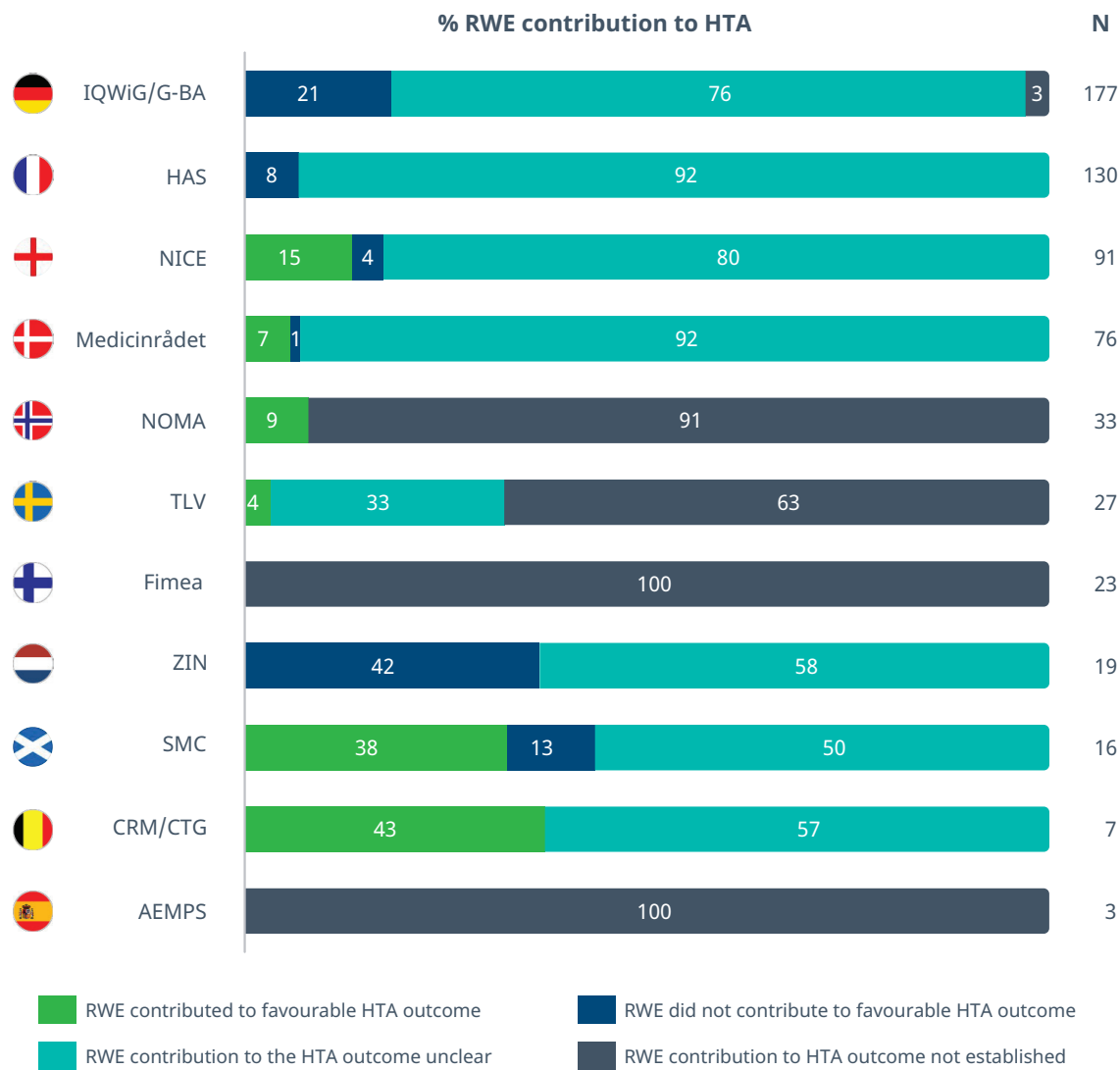
Nevertheless, ECA remains the use case in which RWE most frequently contributed to HTA decision-making overall, with RWE playing an important role in more than half of all assessments. This indicates that further strengthening methodological standards and analytical rigour for RWE used in ECAs could offer the greatest potential to influence HTA outcomes.

Rather than discouraging the use of RWE, recent trends highlight a critical need for robust methodological standards when designing and presenting ECAs. Historically, some submissions have lacked sufficient rigour, which limited their acceptance. However, even when submission of ECAs did not contribute to a positive HTA outcome, its contribution should be judged relative to a scenario where no RWE was submitted. Across most HTA bodies, the preference

is clear: ECAs using IPD from RWE are generally viewed more favourably than naïve comparisons or unanchored MAICs.³⁷ High-quality RWE, developed transparently and contextualized appropriately, can significantly strengthen the credibility of comparative analyses and support informed decision-making (as illustrated in the case studies discussed later).

Additionally, in rare cases (N=1), RWE used to justify unmet need/burden of illness have also not contributed to a positive HTA outcome (Figure 3). This likely reflects situations where HTA bodies themselves used RWE to argue a lack of unmet need, contrary to HTD claims. These cases highlight that RWE can be used not only by HTDs, but also by HTA bodies during their appraisals.

Figure 4. RWE contribution to HTA outcomes per HTA body (2021-2024)



Receptiveness to RWE across HTA bodies

The majority of HTA submissions including RWE were identified within evaluations conducted by IQWiG/ Federal Joint Committee (G-BA), totalling 177 cases (**Figure 4**). Among these, 21% (N=37) were cases where RWE did not contribute to favourable HTA outcomes, primarily when RWE was used as an ECA, leading to a 'no added benefit' decision. Importantly, these outcomes would likely have been negative irrespective of the inclusion of RWE, reflecting the absence of robust head-to-head clinical trial evidence.

This underscores the particularly high evidentiary bar applied by IQWiG/G-BA for comparative claims and highlights the need for rigorous methodological standards when ECAs are used (as seen in case studies discussed later). Encouragingly, most RWE submissions which contributed to a favourable HTA outcome were seen in the UK (NICE, SMC), Belgium (Commission for Reimbursement of Medicines (CRM/CTG)) and in the Nordics (Denmark, Sweden, Finland).



Case studies: Use of RWE as an External Control Arm — Libtayo and Ayvakit

To explore the practical challenges and opportunities of using RWE in oncology HTAs, we examined the HTAs of Libtayo and Ayvakit, which were both assessed based on single-arm trials (SATs) and relied on the use of retrospective studies as external controls to establish their effectiveness versus current SoC.

Libtayo (cemiplimab) in locally advanced/metastatic cutaneous squamous cell carcinoma (cSCC) - resubmission			
RWE purpose	Comparative Effectiveness: External Control Arm for R2810-ONC-1540 uncontrolled Phase 2, open-label, multicentre trial		
HTA body	 IQWiG/G-BA	 HAS	 NICE
HTA outcome	No added benefit	ASMR IV (3rd submission)	Positive with restrictions
RWE source and approach	• Unadjusted (naïve) comparison of individual arms from Phase 2 trial (R2810-ONC-1540) and Hillen et al. 2018⁴¹ • Hillen et al. 2018⁴¹ a retrospective, non-interventional cohort study by the Dermatologic Cooperative Oncology Group (DeCOG) of data on patients with newly-diagnosed advanced cSCC treated in 24 centres in Germany and Austria between January 2010 and December 2011	• TOSCA RWE study: Indirect comparison between the data of patients treated with Libtayo as part of its ATU cohort and a historical cohort of patients treated in France in the same expert centers by systemic off-label treatments	• Inverse Probability of Treatment Weighting (IPTW) and naïve comparison vs UK retrospective chart review (platinum-based chemotherapy (ChT)) • Unanchored MAIC, STC and naïve vs ChT (Jarkowski et al. 2016)⁴² and vs Best-Supportive Care (BSC) (Sun et al. 2019)⁴³ • Fitting survival directly to observed Kaplan-Meier data from Sun et al. 2019, the UK retrospective chart review, and Jarkowski et al. 2016⁴⁰
RWE contribution	RWE did not contribute to favourable HTA outcome	RWE contributed to favourable HTA outcome	RWE contributed to favourable HTA outcome
Key critique	• Overall survival (OS) follow-up definition differed between trial and RWE • No comparison of patient-relevant outcomes symptoms, quality of life (QoL) and safety, but benefit assessment requires evidence on risks/harms • Unanchored ITCs only acceptable if there are dramatic effects, which was not the case here • Propensity based methods needed but drug developer claims to have no comparator IPD despite submitting unpublished subgroup • No overlap between populations in some characteristics	• OS follow-up definition differed between trial and RWE • No comparison of patient-relevant outcomes symptoms, QoL and safety • The ITC was restricted by measurement / selection bias and limited follow-up • However, evidence was sufficient to demonstrate benefit for a positive evaluation	• Chart review lacked face validity due to not overlapping with pivotal trial in key characteristics and not using relevant comparator (BSC) • Sun et al. 2019⁴³ was accepted because it assessed BSC, and despite being from the US and only 20 patients were relevant for the ITC

Ayvakit (avapritinib) in Advanced Systemic Mastocytosis (AdvSM)

RWE purpose	Comparative Effectiveness: External Control Arm for PATHFINDER and EXPLORER single arm studies		
HTA body	 IQWiG/G-BA	 HAS	 NICE
HTA outcome	Hint for a non-quantifiable additional benefit	ASMR IV	Positive recommendation
RWE source and approach	• BLU-285-2405 study (PATHFINDER and EXPLORER studies (pooled) versus best available therapy (BAT) cohort), a propensity score (PS)-adjusted indirect comparison without a bridge comparator for the indication of AdvSM • Pilkington et al. 2022⁴⁴ (external control study, indirect comparison), a publication on an indirect comparison of avapritinib with midostaurin in patients with advanced systemic mastocytosis	• Indirect comparison to compare the results of avapritinib from the single arm trials PATHFINDER and EXPLORER with BAT for patients with AdvSM • It uses individual data from retrospectively collected medical records of adult patients who have received systemic therapy (any line) since 1 Jan 2009 in UK, Spain, Austria, Germany, and US	Comparative data was derived from two sources: • IPW analysis using BLU-285-2405, real-world data retrospective data on BAT • MAIC in which individual patient data from PATHFINDER and EXPLORER were indirectly compared with aggregate data from two Phase 2 studies evaluating midostaurin in people with AdvSM (D2201 and A2213)
RWE contribution	RWE contribution to the HTA outcome unclear	RWE contributed to favourable HTA outcome	RWE contributed to favourable HTA outcome
Key critique	• Considerable uncertainties mainly due to the lack of a systematic literature search and assessment to identify confounders • Missing data with insufficient imputation leading to insufficient confounder adjustment	• Despite criticizing that the propensity score model in the ITC did not include all relevant prognostic factors, HAS was willing to acknowledge that the ITC suggested a benefit of Ayvakit over the ECA — leading to it providing a partial response to the unmet need • HAS also took into account how poor the evidence for any of the clinically relevant comparators were and that this indication thus has a high unmet need	• Disagreements about confounder selection and not estimating bias of missing confounder adjustment • The condition is rare and there would be difficulties in the collection of the data justifying uncertainties

Tailoring RWE to local HTA expectations

Both case studies underscored several critical considerations for the effective use of RWE. First, a “one-size-fits-all” approach is insufficient as each HTA body applies distinct evidentiary standards, requiring tailored RWE strategies that align with local methodological expectations and decision frameworks.

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In practice, the HTDs adapted their approach for each HTA body: NICE valued UK-specific chart reviews, HAS preferred French ATU data, G-BA dismissed non-German cohorts. This underscores the need for greater harmonization on the one hand, and generalisability to the respective authorities on the other. In response to this challenge, the GetReal Institute is developing a framework³⁸ reviewing RWD-based ECA methodologies and stakeholder guidance with the aim of consolidating and clarifying expectations for ECA studies using RWD.

Limitations in long-term follow-up

The absence of long-term follow-up data in RWE can significantly limit its utility in demonstrating durability of response and survival outcomes, which are often pivotal in oncology assessments. Without robust longitudinal data, HTA bodies may question the sustainability of clinical benefit.

Gaps in PROs

The lack of comparative data on patient-relevant outcomes, such as symptoms and quality of life, can undermine the perceived added benefit of a

therapy. Therefore, the inclusion of validated PRO instruments in both clinical trials and RWE studies is essential to strengthen the evidence base and address HTA expectations.

Methodological rigour

Rigorous methodological planning is key. Confounders should be systematically identified a priori and ECAs should ideally be pre-specified within trial protocols.

RWE datasets must be as complete as possible with respect to key prognostic variables. Where data are missing, sensitivity analyses using imputation should be conducted, with conservative estimates clearly justified, particularly in rare diseases where data limitations are common.

In the UK context, ECAs are best reserved for scenarios where unanchored ITCs (i.e., MAICs, STCs) with SAT data are unfeasible, for example due to a small effective sample size post-matching/weighting in the case of MAICs, or model misspecification due to a skewed covariate distribution in the case of STCs.

It is evident that rigorous methodological planning is key to the successful use of RWE in HTA. ECAs therefore remain a viable and valuable option in HTA submissions, provided they are designed and presented in a way that meets the evidentiary standards of the relevant HTA body. When methodological expectations are met, ECAs can contribute meaningfully to positive HTA outcomes, especially in settings where traditional RCTs are impractical or insufficient.

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Case study: Use of RWE in Carvykti, Tecvayli and Talvey HTA submissions

We next examined the HTA evaluations of Carvykti, Tecvayli, and Talvey in multiple myeloma. We focused on how data from LocoMMotion, a prospective, non-interventional study, was used across the three submissions, as this is one of the few prospective external control arm studies submitted for HTA thus far.

Carvykti (Ciltacabtagene autoleucel) in 4L+ r/r multiple myeloma		
RWE purpose	Comparative Effectiveness External Control Arm for CARTITUDE-1 single-arm study	
HTA body	 IQWiG/G-BA	 HAS
HTA outcome	(2023) Non-quantifiable additional benefit	(2022) ASMR V
RWE source and approach	• LocoMMotion, a prospective, non-interventional study, was used by the drug developer for an indirect comparison of ciltacabtagene autoleucel versus standard therapy in everyday care • Indirect comparison without a common comparator between data from the single-arm, open-label phase Ib/II study CARTITUDE-1 with data from the prospective, non-interventional LocoMMotion study in the dossier • Furthermore, as part of the written comment procedure, a pooled analysis of the phase Ib/II CARTITUDE-1 and phase 3 CARTITUDE-4 studies in comparison to the LocoMMotion study based on the individually collected patient data was presented	• Indirect comparisons based on propensity scores estimated on individual data: comparison of Carvykti to the LocoMMotion, EMMY, FLATIRON and MAMMOTH cohorts • MAIC-type indirect comparisons on aggregated data: comparison of Carvykti with the 3 clinically relevant comparators: Abecma (idecabtagene vicleucel), Blenrep (belantamab mafodotin) and Nexpovio (selinexor)
RWE contribution	RWE contribution to the HTA outcome unclear	RWE contribution to the HTA outcome unclear
Key critique	• Inclusion/exclusion criteria of LocoMMotion study do not fully reflect the medical suitability and selection mechanisms of CAR-T cell therapy • High number of missing values for some essential baseline characteristics and potential confounders • Different collection times and methods of the baseline characteristics	• Possible selection bias linked to the definition of the initial time • Heterogeneity in the assessment of certain endpoints (response, progression) • Heterogeneity of strategies for taking prognostic factors and potential effect modifiers into account • Uncertainty on the balance obtained after weighting of the treatment groups • Violation of the common support hypothesis
Notes	• Janssen withdrew its evidence submission to NICE	

Tecvayli (teclistamab) in 4L+ r/r multiple myeloma		
RWE purpose	Comparative Effectiveness: External Control Arm for MajesTEC-1 open-label Phase Ib/II study	
HTA body	 HAS	 NICE
HTA outcome	(2024) ASMR V	(2024) Positive recommendation
RWE source and approach	Three indirect comparisons between patients in the MajesTEC-1 open-label Phase Ib/II study and - Individual patient data from a prospective multicenter LocoMMotion cohort - Individual patient data from the US FLATIRON database - Individual long-term follow-up data from patients exposed to daratumumab included in the POLLUX, CASTOR and EQUULEUS studies <i>Here, we focus on the first indirect comparison vs LocoMMotion cohort</i>	Indirect treatment comparison using Individual Patient Data (IPD) from the UK RW TCE (triple class exposed) cohort study and MajesTEC-1 open-label Phase Ib/II study, utilising an inverse probability of treatment weighting (IPTW) approach (in base case), resulting in a MajesTEC-1 cohort that was highly matched to the UK RW TCE population
RWE contribution	RWE did not contribute to favourable HTA outcome	RWE contributed to favourable HTA outcome
Key critique	- Assessment of the ORR was performed by a non-independent review committee for the LocoMMotion cohort, and by an IRC for the MajesTEC-1 study. Therefore, measurement bias on this endpoint cannot be ruled out - Uncertainties regarding the specific effect of the treatment, considering the methodological weaknesses of indirect comparisons - Due to methodological limitations, no formal conclusions were drawn from this indirect comparison	- IPTW method may be unstable, estimated treatment effect may be biased - Four priority prognostic factors (cytogenetic profile, ISS stage, time to progress on last regimen, extramedullary plasmacytoma) not adjusted - Violation of proportional hazard assumption
Notes	- No RWE was submitted to IQWiG/G-BA - LocoMMotion was not used in the NICE submission	

Talvey (talquetamab) in 4L+ r/r multiple myeloma

RWE purpose	Comparative Effectiveness: External Control Arm for Phase I/II MonumenTAL-1 non-comparative, multi-cohort, multicenter study	
HTA body	 HAS	 NICE
HTA outcome	(2024) ASMR V	(2025) Positive with restrictions
RWE source and approach	Indirect Treatment Comparison between LocoMMotion and Phase I/II MonumenTAL-1 non-comparative, multi-cohort, multicenter study	Indirect treatment comparison of talquetamab and teclistamab using 2 retrospective, non-interventional real-world studies , REALiTEC (teclistamab; n=113) and REALiTAL (talquetamab; n=93), evaluating real-world outcomes in people with triple-class exposed relapsed or refractory multiple myeloma and included people from the UK
RWE contribution	RWE contribution to the HTA outcome unclear	RWE contributed to favourable HTA outcome
Key critique	• Grouping of data coming from different sources • Choice of real-life comparators very diverse, without data on Tecvayli or Elrexfio • Important criteria, e.g., cytogenetic risk or the number of neutrophils, not taken into account • Residual differences in the measured effect modifiers (SMD>0.10) • Confounding bias on unmeasured factors • Multiple criteria, without controlling the risk of error	• High level of uncertainty associated with the OS hazard ratio for talquetamab compared with teclistamab, including » Early separation of the talquetamab and teclistamab OS curves » Lack of correlation between progression-free survival (PFS) and OS, particularly the size of the overall-survival benefit » Both REALiTEC and REALiTAL were done after the COVID-19 pandemic, while the trials were conducted during • However, the broad range of OS hazard ratio estimates from different analyses helped to characterise the uncertainty
Notes	• No RWE was submitted to IQWiG/G-BA • LocoMMotion was not used in the NICE submission	

Methodological rigour

Robust data integrity and methodological rigour are essential for the acceptance of RWE in HTA. First, ensuring data completeness is critical. High levels of data missingness can undermine the validity of comparative analyses and introduce bias. HTDs should systematically report the extent and pattern of missing data and pre-specify handling strategies within study protocols and Statistical Analysis Plans (SAPs). This includes the use of imputation techniques and sensitivity analyses. Transparent reporting enhances credibility and reduces the risk of bias in comparative analyses.

Meaningful comparisons across trial arms and external cohorts require harmonisation of baseline data collection, both in terms of timing and methodology. In the Carvykti HTA submissions, differences in baseline definitions between the trial and RWE cohorts undermined the validity of indirect comparisons. Similarly, heterogeneity in the assessment of certain endpoints was flagged by HAS. Employing blinded or independent review committees can mitigate measurement bias and strengthen the reliability of outcome evaluations.

Addressing residual confounding

HTA bodies expect rigorous methodological planning when RWE is used as an external comparator. In Carvykti's submission, residual confounding persisted despite matching, with Standardised Mean Differences (SMDs) between baseline characteristics of the trial and the ECA exceeding acceptable thresholds. As mentioned, to address this, HTDs should strive to pre-specify ECAs in the trial protocol to enhance transparency, reduce bias, and align with HTA expectations for robust evidence generation.

Where indirect comparisons are necessary due to small sample sizes, methods such as MAICs and STCs are preferred, provided they are transparently executed and supported by robust baseline and follow-up data.

In settings such as 4L+ r/r multiple myeloma, RWE can play a critical role in contextualizing clinical benefit. However, its contribution is contingent on

methodological robustness and alignment with HTA expectations. These practices collectively enhance the transparency, reproducibility, and scientific credibility of RWE, thereby increasing its utility in HTA decision-making.

Discussion

The use of RWD in HTA submissions has seen mixed levels of acceptance across different HTA agencies, however, its role is steadily expanding and gaining recognition as a valuable complement to traditional evidence sources. While acceptance varies, there is a clear trend toward greater integration of RWE, particularly in Cost-Effectiveness (CE)-focused evaluations compared to those prioritizing clinical benefit assessment.

HTA bodies with a strong focus on CE (e.g., NICE) tend to be more receptive to RWE, leveraging it not only to support economic modelling but increasingly to contextualize outcomes and reduce uncertainty in early or immature trial data.⁴⁴ For example, in the case of Libtayo, ECAs derived from RWD were successfully used for long-term extrapolation of OS curves in the absence of mature trial data.

In contrast, HTA bodies prioritising clinical effectiveness, e.g., G-BA or HAS, are sceptical of ECAs.⁴⁴ These agencies place exceptionally high value on randomisation and comparisons to non-randomised data typically face methodological scrutiny. However, this stance is evolving; notable exceptions, such as the cases of Libtayo and Ayvakit, demonstrate that well-designed RWE can positively influence decisions, signaling a tentative shift toward more pragmatic evidence acceptance.

Overall, HTA bodies are becoming more prescriptive and detailed in their guidance on RWE study designs, methodological and procedural requirements, analytic approaches for bias mitigation as well as reporting standards, thus raising the bar for submissions and rewarding RWE that directly addresses decision-critical uncertainties.

Overall, HTA bodies are becoming more prescriptive and detailed in their guidance on RWE study designs, methodological and procedural requirements, analytic approaches for bias mitigation as well as reporting standards, thus raising the bar for submissions and rewarding RWE that directly addresses decision critical uncertainties.

Therefore, HTDs must pay close attention to the design, planning, and operationalization of RWE studies to ensure alignment with agency-specific requirements while pre-empting avoidable methodological criticisms. Common objections include the risk of selection bias, temporal misalignment of data collection, and insufficient control for confounding and effect modification, often stemming from missing data, heterogeneity in patient populations, or inconsistent measurement practices. For instance, IQWiG has acknowledged the potential acceptability of ITCs using RWE but sets a notably high evidentiary threshold: data should be collected prospectively, concurrently with the clinical trial, and use identical variables, measurement schedules, and follow-up protocols. These expectations underscore the growing importance of planning and methodological rigor to unlock RWE's potential.

Beyond methodological rigor, accessibility to high-quality RWD is critical. Initiatives to standardize data and improve interoperability are accelerating, paving the way for more robust and credible RWE.

Another positive trend is the recognition that early planning of ECAs by HTDs can transform RWE from a reactive tool into a strategic asset. When incorporated during trial design, RWE strengthens submissions and reduces uncertainty, positioning HTDs to meet evolving HTA expectations.

In summary, while challenges remain, current trends indicate that RWE is playing an increasingly important, though context-dependent, role in HTA decision-making, with its influence varying by use case, therapeutic area, and agency.

Another positive trend is the recognition that early planning of ECAs by HTDs can transform RWE from a reactive tool into a strategic asset. When incorporated during trial design, RWE strengthens submissions and reduces uncertainty, positioning HTDs to meet evolving HTA expectations.

Conclusion

Real-world evidence is assuming an increasingly significant role in HTA decision-making. As HTA bodies evolve their frameworks to accommodate broader evidence sources, HTDs must move beyond ad hoc inclusion of RWE and instead adopt a **strategic, proactive approach** to its generation and use. This means developing a clear plan for how RWE will be leveraged, **tailored to the evidentiary needs and methodological expectations of each HTA body**.

Equally important is the need for more harmonised guidance across HTA agencies to reduce variability in evidentiary standards and methodological requirements. As HTA bodies provide clearer direction on acceptable data sources, study designs, and analytic approaches, greater internal HTA alignment will be essential to facilitate the design of RWE strategies that meet multi-country expectations without duplicating effort, delaying access, and inhibiting innovation.

To improve the acceptability of RWE in HTA submissions, HTDs should:

- **Seek early scientific advice from HTA bodies** to ensure that planned RWE approaches are acceptable and aligned with local decision frameworks
- **Strive to pre-specify ECAs within trial protocols** and adopt the estimand and target trial emulation frameworks
- **Ensure data completeness**, systematically reporting missing data patterns and pre-specifying imputation strategies in statistical analysis plans
- **Harmonise baseline data collection across cohorts** in terms of timing and methodology to enable meaningful comparisons
- **Maintain consistency in endpoint assessment**, ideally using blinded or independent review committees to mitigate measurement bias
- **Align RWE generation with local clinical practice and HTA expectations**, tailoring comparator cohorts to reflect country-specific standards of care
- **Transparently document all methodological choices**, including assumptions, adjustments, and sensitivity analyses, to enhance credibility and reproducibility

When developed with methodological rigor and aligned to local HTA standards, RWE can be a powerful and viable component of the evidence package. With early engagement and thoughtful planning, RWE can meaningfully support positive HTA outcomes and potentially pricing negotiations by helping to address evidence gaps that traditional trials may leave unfilled.

For guidance on optimizing your RWE approach early to maximize HTA success, including questions related to external control arms, please reach out to the [IQVIA Real-World Evidence team](#).

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